

# Ajay Kumar Saini

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## SUMMARY

Full-Stack Engineer who builds and runs a 147,000-line production ERP serving seven business modules on Google Cloud Run. Works day to day on distributed background jobs, safe data sync, and large third-party API integrations, backed by 2,856 automated tests.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, SQL, C++, C

**Backend:** Node.js, Next.js, REST APIs, WebSockets, Mongoose ODM, Prisma ORM, NextAuth, JWT, RBAC

**Databases:** MongoDB, PostgreSQL, Schema Design, Indexing, Aggregation Pipelines, Query Optimization

**Frontend:** React, Tailwind CSS, Zustand, Redux Toolkit, shadcn/ui, Server Components

**Cloud & DevOps:** GCP (Cloud Run, Cloud Storage, Vertex AI), AWS, Docker, GitHub Actions, CI/CD, Git

**Integrations & AI:** Amazon SP-API, Zoho Books, WhatsApp Cloud, Google Drive, Gmail, Groq, OAuth 2.0

**Engineering:** System Design, Distributed Locking, Idempotency, Background Workers, Rate Limiting

## EXPERIENCE

### METNMAT Innovations Pvt. Ltd.

on-site / Howrah, West Bengal

Full-Stack Developer Intern

Jan 2026 – Aug 2026

- Owned a 147,000-line production ERP spanning 461 REST endpoints, 60 MongoDB models, and 267 React components across seven business modules, improving data processing efficiency by 15%.
- Designed and implemented a robust, fail-closed upload pipeline with 16 security modules, migrating 238 document binaries to private Google Cloud Storage and shrinking database by 82%.
- Connected Amazon SP-API, Gmail, and Zoho Books through 18 scheduled jobs that reconcile orders, settlements, and GST into double-entry ledgers, cutting manual bookkeeping effort by 70%.
- Developed a quotation-to-invoice engine over five templates with atomic numbering and PDF/DOCX export, cutting document turnaround by 50%, and Introduced a WhatsApp AI assistant across 22 read-only tool handlers.
- Raised automated coverage to 2,856 unit and integration tests across 343 suites, and Streamlining every release to Google Cloud Run inside Docker containers.

### Indian Institute of Technology Jodhpur

Jodhpur, India

Research Intern

May 2025 – Jul 2025

- Simulated hot tearing in Al-206 castings using the CRC criterion, and isolated the critical solid-fraction range of 0.88 to 0.96.
- Validated those predictions against SEM and XRD analysis of real castings, improving casting yield by 8.4%.

## PROJECTS

### Anwasha Festival Platform, IIT Patna Cultural Fest

GitHub

Next.js, PostgreSQL, Node.js, Authentication

Dec 2024 – May 2026

- Launched the registration and events platform for the campus cultural fest, which served more than 3,000 active users.
- Designed a normalized PostgreSQL schema that backs the event catalogue and stores over 3,000 participant registrations.
- Led the web team as Technical Lead, coordinating delivery of the registration flow, authentication, and admin tooling.

### Questivo, AI Assessment Engine with Real-Time Streaming

GitHub

React, Node.js, Express, Prisma, PostgreSQL, WebSockets, Redis, Docker, AWS, Groq SDK

Jan 2026

- Built an LLM assessment engine on the Groq SDK that held session isolation under more than 500 concurrent requests.
- Streamed model output token by token over WebSockets, cutting perceived response latency by 60% compared with the earlier polling approach.
- Decoupled evaluation behind Redis-backed job queues so long LLM calls never blocked the Express REST API, then shipped the service to AWS EC2 and S3 with Docker.

## EDUCATION

### Indian Institute of Technology, Patna

Patna, Bihar

B.Tech in Metallurgical and Materials Engineering | CPI: 7.88/10.0

Jul 2023 – Jul 2027

**Coursework:** Data Structures & Algorithms, DBMS, Machine Learning, Object-Oriented Programming

### Bhagawati Sr. Sec. School (CBSE)

Rajasthan

Class XII, Senior Secondary, 91.4%

2022

## ACHIEVEMENTS & LEADERSHIP

- Selected for the **Data & AI Operations Internship** at Grindwell Norton Limited, part of the Saint-Gobain Group.
- Ranked in the **top 2%** of more than **1.2 million** candidates in JEE Advanced 2023, and solved **650+ algorithmic problems** on LeetCode.
- Served as Head of Sponsorship for **E-Cell and Celesta at IIT Patna**, leading corporate partnerships for the annual fest.
- Certified in Full-Stack AI Engineering (ML Foundations) and Secure Python and Django development through Udemy.